



Vehicle Type Classification Using CNN and HOG-SVM Based on Digital Images

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ABSTRACT

Vehicle image classification is an important task in computer vision because it can support intelligent transportation, visual search, and automated data organization. This study implements and compares two approaches for vehicle type classification: Histogram of Oriented Gradients with Support Vector Machine (HOG-SVM) and Convolutional Neural Network (CNN). The dataset consists of 6,480 vehicle images divided into 10 classes. The preprocessing stage includes image resizing to 128 x 128 pixels, RGB conversion, pixel normalization, label encoding, and stratified train-test splitting with an 80:20 ratio. The HOG-SVM model is used as a classical machine learning baseline, while the CNN model is designed using three convolutional blocks, batch normalization, max pooling, dense layers, dropout, and softmax output. The experimental results show that HOG-SVM obtained an accuracy of 46.29%, while CNN achieved 59.00% accuracy on the test data. A new image prediction test was conducted using ten external vehicle images. The CNN model successfully classified most motorcycle images with confidence values reaching 99.87%, while several four-wheeled vehicle categories produced lower confidence due to visual similarity among classes.

INTRODUCTION

Image-based vehicle classification is one of the practical problems in computer vision. The ability of a system to recognize vehicle categories from digital images can be used in traffic monitoring, parking management, visual documentation, and intelligent transportation applications. In academic contexts, this problem is also useful for studying how machine learning algorithms process visual patterns, because vehicle images contain edges, shapes, colors, shadows, and background variations that can influence classification performance.

The main challenge in vehicle classification is the similarity between several vehicle categories. Classes such as sedan, hatchback, SUV, and MPV may have similar body shapes when captured from

different viewpoints. In addition, motorcycle and bicycle-like vehicle classes can also be difficult to separate when image resolution, lighting, and background conditions are inconsistent. These conditions make classification more complex than simple object recognition on clean images.

Previous studies show that classical feature extraction methods such as Histogram of Oriented Gradients (HOG) can represent object shape by capturing gradient orientation patterns [5]. HOG is commonly combined with Support Vector Machine (SVM), which is effective for supervised classification tasks [6]. However, handcrafted features may not capture complex visual patterns as strongly as deep learning methods. Convolutional Neural Networks (CNNs) have become widely used in image classification because they can learn hierarchical visual features directly from image data [1]-[4].

Based on this problem, this study compares HOG-SVM and CNN for vehicle type classification using a dataset containing 6,480 images across 10 classes. The comparison is intended to show the difference between a classical machine learning approach and a deep learning approach in the same dataset and preprocessing setting. The results are expected to provide a clear explanation of how feature extraction strategy affects classification performance.

The objective of this study is to implement a complete vehicle classification workflow starting from dataset checking, preprocessing, model training, evaluation, model saving, and new image prediction. The novelty of this work is not the proposal of a new algorithm, but the implementation and empirical comparison of HOG-SVM and CNN in a reproducible student-level machine learning project using Python-based tools.

MATERIALS AND METHODS

Research Design

This research uses an experimental quantitative approach. The experiment focuses on building and evaluating classification models for vehicle images. Two methods are compared: HOG-SVM as a classical machine learning baseline and CNN as a deep learning model. Both methods use the same dataset split to keep the comparison consistent.

The implementation was carried out using Python 3.12 in a local computer environment. The main libraries used include OpenCV for image reading and preprocessing, NumPy for numerical array handling, Scikit-learn for label encoding, train-test splitting, SVM modeling, and evaluation, Scikit-image for HOG feature extraction, and TensorFlow/Keras for CNN training [8]-[13].

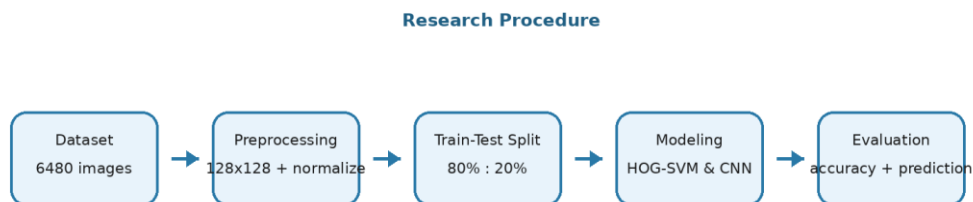


Figure 1. Research Procedure of Vehicle Classification

Data and Instruments

The dataset used in this study contains 6,480 vehicle images with labels stored in a CSV file. The labels consist of 10 vehicle classes. Each image is read from the training folder based on its ID, then paired with its corresponding label. The dataset was split into 5,184 training images and 1,296 testing images using an 80:20 ratio.

The instruments used in this research are Python scripts arranged according to the workflow. The file `cek_dataset.py` is used to check dataset availability and class distribution. The file `preprocessing.py` processes images, encodes labels, splits the dataset, and saves the processed arrays into `dataset_preprocessed.npz`. The files `hog_svm.py` and `train_model.py` are used for model training, while `predict.py` is used for new image prediction.

Table 1. Dataset and Preprocessing Summary

Aspect	Description
Total data	6,480 vehicle images
Number of classes	10 vehicle classes
Training data	5,184 images (80%)
Testing data	1,296 images (20%)
Image size	128 x 128 pixels
Preprocessing	RGB conversion, normalization, label encoding, stratified train-test split

Research Procedure

The research procedure begins with dataset checking to ensure that images and labels are correctly matched. After that, preprocessing is performed by resizing images to 128 x 128 pixels, converting BGR images to RGB, normalizing pixel values into the range 0-1, and encoding text labels into numerical labels. The processed dataset is saved in compressed NPZ format to avoid repeated image loading during later experiments.

For the HOG-SVM method, each RGB image is converted into grayscale using the average of its three color channels. HOG features are extracted using 9 orientations, 8 x 8 pixels per cell, 2 x 2 cells per block, and L2-Hys normalization. The extracted features are then classified using an SVM model with a linear kernel.

For the CNN method, the model architecture consists of three convolutional layers with 32, 64, and 128 filters. Each convolutional layer is followed by batch normalization and max pooling. The extracted feature maps are flattened and passed into a dense layer with 256 neurons, followed by dropout with a rate of 0.5 and a final softmax layer for 10-class classification. The model uses Adam optimizer and sparse categorical cross-entropy loss.

Table 2. Model Configuration

Model	Configuration
HOG-SVM	HOG orientations = 9, pixels per cell = 8 x 8, cells per block = 2 x 2, block norm = L2-Hys, SVM kernel = linear
CNN	Conv2D 32, 64, 128 filters; BatchNormalization; MaxPooling; Flatten; Dense 256; Dropout 0.5; Softmax output

Data Analysis Techniques

Model performance is analyzed using accuracy as the main evaluation metric. Accuracy is calculated by comparing the number of correct predictions with the total number of test samples. The HOG-SVM result is also supported by a confusion matrix to observe prediction distribution across classes. For CNN, the test accuracy and new image prediction confidence are used to evaluate whether the trained model can classify images outside the training data.

The accuracy formula used in this study is shown in (1), where TP represents correct predictions and N represents the total number of tested samples.

$$Accuracy = \frac{\text{Correct Predictions}}{\text{Total Test Samples}} \times 100$$

RESULTS AND DISCUSSION

Results

The dataset checking and preprocessing stages were completed successfully. A total of 6,480 images were processed and divided into 5,184 training images and 1,296 testing images. The preprocessing output was saved as `dataset_preprocessed.npz`, and the label encoder was saved as `label_encoder.pkl`. This made the training stage faster because the image dataset did not need to be read repeatedly from the original image folder.

The HOG-SVM method achieved an accuracy of 46.29% on the test data. Meanwhile, the CNN model achieved an accuracy of 59.00%. The CNN model also generated a trained model file (`model_kendaraan.keras`) for inference. Ten external vehicle images were tested to evaluate the model's generalization capability. The model correctly recognized most motorcycle images with confidence values reaching 99.87%, whereas some four-wheeled vehicles produced lower confidence because several vehicle classes shared similar visual characteristics

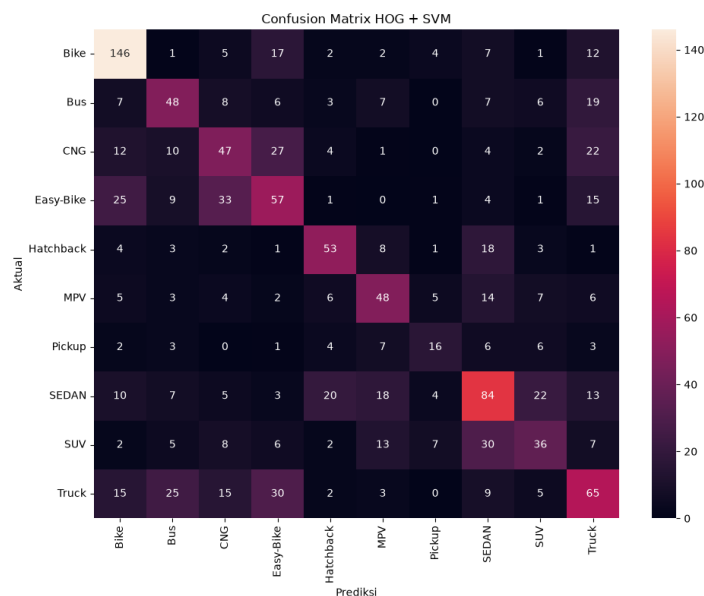


Figure 2. Confusion Matrix of HOG-SVM

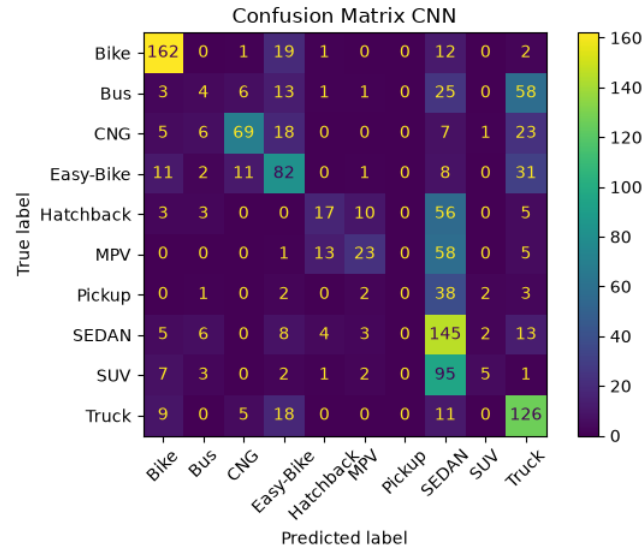


Figure 3. Confusion Matrix of CNN

The confusion matrix shows that the HOG-SVM model correctly classified several vehicle categories, although confusion frequently occurred among visually similar classes such as Easy-Bike, Truck, and CNG.

The CNN confusion matrix demonstrates better classification performance than HOG-SVM, particularly for Bike, Sedan, and Truck categories. However, several misclassifications still occurred among MPV, SUV, and Sedan due to their similar visual characteristics.

Table 3. Accuracy Comparison of HOG-SVM and CNN

Method	Accuracy	Output
HOG-SVM	46.29%	Confusion matrix and classification report
CNN	59.00%	Saved model: model_kendaraan.keras

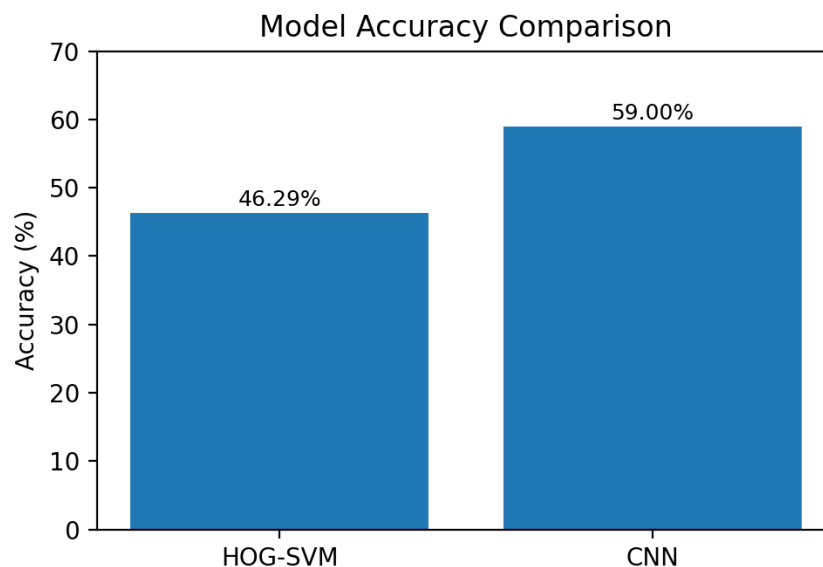


Figure 4. Accuracy Comparison Between HOG-SVM and CNN

Table 4. New Image Prediction Result

Test Image	Predicted Class	Confidence
	Bike	95.91%
	Easy-Bike	59.56%
	SEDAN	31.39 %
	Bike	90.58 %

	SEDAN	51.74 %
	Bike	99.87 %
	Bike	99.64 %

	MPV	30.97 %
	Bike	97.9 %
	SEDAN	90.96 %

Discussion

The result shows that CNN performed better than HOG-SVM by approximately 12.71 percentage points. This improvement is reasonable because CNN learns visual features automatically through convolutional filters, while HOG-SVM depends on handcrafted gradient features. HOG can represent object edges and shape, but it may lose color and contextual information that helps distinguish visually similar vehicle classes.

Although CNN achieved higher accuracy, the result is still moderate. Several factors may affect the performance. First, the dataset contains 10 classes with similar visual characteristics, especially among car body types. Second, vehicle images may have different viewpoints, backgrounds, lighting, and scale. Third, the CNN architecture used in this study is intentionally simple to fit a local training environment, so it may not capture features as strongly as deeper pretrained models such as VGG, ResNet, or Inception [3], [4].

The new image prediction results indicate that the trained CNN model can be used for basic image classification outside the training dataset. Most motorcycle images were correctly classified as Bike with confidence values above 90%, reaching a maximum of 99.87%. This indicates that the model learned motorcycle visual features effectively, although confidence for several four-wheeled vehicle classes remained lower because of visual similarity among categories. The confusion matrix indicates that the CNN model correctly classified most Bike and Truck images, while confusion frequently occurred among Bus, SUV, MPV, and Hatchback due to their similar visual characteristics. This suggests that CNN is able to capture dominant visual patterns, although several vehicle categories remain difficult to distinguish.

The prediction confidence varies depending on the similarity between the input image and the training data. Images with clear visual characteristics, such as motorcycles, generally produce higher confidence values, while sedan images with more diverse backgrounds or viewing angles may result in lower confidence despite being classified correctly.

The external prediction results also show that the CNN model performs consistently on several motorcycle images, with confidence values above 90%, while four-wheeled vehicles produce more varied confidence scores. For example, the Range Rover image was classified as MPV with 30.97% confidence because SUVs and MPVs share similar body proportions and visual characteristics. This indicates that the model has learned dominant motorcycle features more effectively than visually similar car categories.

Research Implications

Practically, this implementation can be used as a basic prototype for vehicle image classification. The project demonstrates the complete workflow from dataset preparation to model inference. The use of a saved CNN model also makes the system suitable for further development into a desktop, web, or mobile-based prediction application.

Theoretically, this study supports the common finding that deep learning models are generally more suitable for image classification than handcrafted feature methods when sufficient data are available. However, the performance also shows that model architecture, dataset quality, and training strategy strongly influence final accuracy.

The limitation of this research is that the CNN architecture is still relatively simple and trained without transfer learning. Future research may improve performance by using data augmentation, pretrained models, hyperparameter tuning, balanced datasets, and more detailed metrics such as precision, recall, F1-score, and per-class confusion matrix analysis.

CONCLUSION

This study successfully implemented a vehicle type classification workflow using HOG-SVM and CNN. The dataset consisted of 6,480 vehicle images from 10 classes and was divided into 5,184 training images and 1,296 testing images. The HOG-SVM method achieved 46.29% accuracy, while the CNN

model achieved 59.00% accuracy. The result shows that CNN performed better because it can learn visual features automatically from image data.

The project also demonstrated model inference using a new image, where the trained CNN predicted several motorcycle images with confidence values up to **99.87%**. The limitation of this research is the use of a relatively simple CNN architecture and limited training optimization. Future work can improve the result by applying data augmentation, transfer learning, deeper CNN architectures, and more detailed per-class evaluation metrics.

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